

COSC 594/650

Slater's Condition and Sensitivity

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Lagrangian duality and KKT conditions

$$\begin{array}{ll}\underset{x}{\text{minimize}} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & x \in X\end{array}$$

■ LAGRANGIAN AND DUAL FUNCTION

■ WEAK AND STRONG DUALITY

■ KARUSH–KUHN–TUCKER CONDITIONS

Problem	Weak duality	Strong duality	KKT conditions
General	$d^* \leq p^*$	may fail	necessary if $d^* = p^*$
Convex + Slater	$d^* \leq p^*$	$d^* = p^*$	necessary & sufficient

QUESTION: Why does Slater's condition give strong duality?

Epigraph form of constrained optimization

$$\begin{array}{ll}\underset{x}{\text{minimize}} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & x \in X\end{array}$$

$\Updownarrow$

$$\begin{array}{ll}\underset{x,t,u}{\text{minimize}} & t \\ \text{subject to} & f_0(x) \leq t \\ & f_i(x) \leq u_i, \quad i = 1, \dots, m \\ & u_i \leq 0, \quad i = 1, \dots, m \\ & x \in X\end{array}$$

Geometric interpretation of duality

Set of values: $\mathcal{A} = \{(u, t) \mid \exists x \in X, f_i(x) \leq u_i, f_0(x) \leq t\}$

Convex set: $\mathcal{A}$ is convex if $f_0, f_1, \dots, f_m$ and X are convex

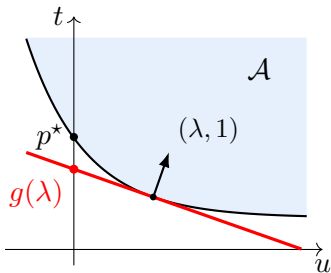
■ PRIMAL PROBLEM

$$p^* = \inf \{t \mid (u, t) \in \mathcal{A}, u \leq 0\}$$

■ DUAL FUNCTION

$$g(\lambda) = \inf \{t + \lambda^\top u \mid (u, t) \in \mathcal{A}\}$$

- ★ $g(\lambda) = t + \lambda^\top u$ – supporting hyperplane
- ★ $(\lambda, 1)$ for $\lambda \geq 0$ – non-vertical normal
- ★ $g(\lambda) \leq p^*$ – intercept on the t -axis



Slater's condition and supporting hyperplanes

Slater point: $(\bar{u}, \bar{t}) \in \mathcal{A}, \quad \bar{u} < 0$

■ SUPPORTING HYPERPLANE

$$a^\top u + \beta t \geq \beta p^*, \quad \forall (u, t) \in \mathcal{A}$$

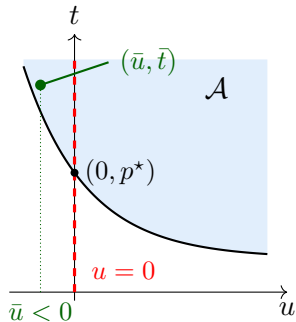
$$a \geq 0, \quad \beta \geq 0, \quad (a, \beta) \neq 0$$

■ VERTICAL CASE: $\beta = 0$

$$a \geq 0, \quad a \neq 0 \quad \implies \quad 0 \leq a^\top \bar{u} < 0$$

by contradiction
 $\implies$

$$\beta > 0$$



Slater's condition and strong duality

Dual multiplier: $\lambda^* = \frac{a}{\beta} \geq 0, \quad \beta > 0$

■ SUPPORTING HYPERPLANE

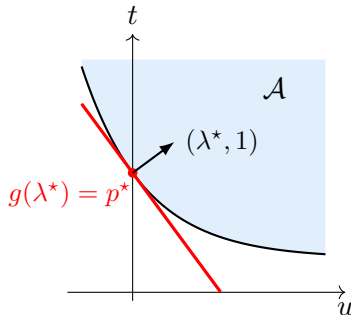
$$t + \lambda^{*\top} u \geq p^*, \quad \forall (u, t) \in \mathcal{A}$$

■ DUAL FUNCTION

$$g(\lambda^*) = \inf \left\{ t + (\lambda^*)^\top u \mid (u, t) \in \mathcal{A} \right\} \geq p^*$$

★ $g(\lambda^*) \leq d^* \leq p^*$ – weak duality

Strong duality: $g(\lambda^*) = d^* = p^*$



The perturbed value function

Strong duality: λ^* dual optimal for $u = 0$

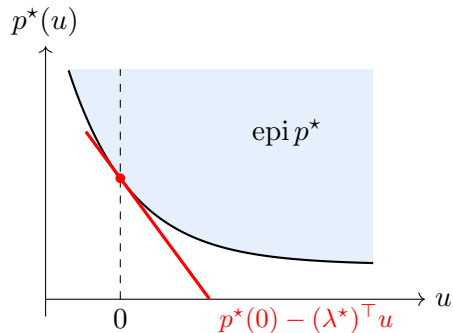
$$\begin{aligned} p^*(u) &:= \underset{x}{\text{minimize}} && f_0(x) \\ &\text{subject to} && f_i(x) \leq u_i, \quad i = 1, \dots, m \\ &&& x \in X \end{aligned}$$

■ GLOBAL LOWER BOUND

$$p^*(0) = g(\lambda^*) \leq L(x, \lambda^*) \leq f_0(x) + \lambda^{*\top} u$$

★ $x \in X, \quad f_i(x) \leq u_i$ – feasible for $p^*(u)$

$$\implies p^*(u) \geq p^*(0) - (\lambda^*)^\top u$$



Multipliers are shadow prices

Global lower bound: $p^*(u) \geq p^*(0) - \lambda^{*\top} u$

■ GLOBAL SENSITIVITY

- ★ large λ_i^* , $u_i < 0$ – tightening greatly increases p^*
- ★ small λ_i^* , $u_i > 0$ – loosening does not decrease p^* much
- ★ $f_i(x^*) < 0 \implies \lambda_i^* = 0$ – inactive constraints are free

■ LOCAL SENSITIVITY

$$\lambda_i^* = - \frac{\partial p^*(0)}{\partial u_i}$$

λ_i^* : price of the i -th constraint

Equality constraints

$$\begin{array}{ll}\underset{x}{\text{minimize}} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p \\ & x \in X\end{array}$$

■ LAGRANGIAN

$$L(x, \lambda, \mu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \mu_i h_i(x)$$

■ DUAL PROBLEM

$$d^* = \max_{\lambda \geq 0, \mu \in \mathbb{R}^p} g(\lambda, \mu), \quad g(\lambda, \mu) = \inf_{x \in X} L(x, \lambda, \mu)$$

★ $x \in \text{relint } X, \quad f_i(x) < 0, \quad h_i(x) = 0$ – Slater point

Convex f_0, f_i + **affine** h_i + **Slater point** $\implies d^* = p^*$

KKT conditions with equality constraints

Differentiable functions: $(f_0, f_1, \dots, f_m)$ and $(h_1, \dots, h_p)$ over $X = \mathbb{R}^n$

Zero duality gap: (x^*, λ^*, μ^*) primal and dual optimal

■ KKT CONDITIONS

- ★ $f_i(x^*) \leq 0, \quad h_j(x^*) = 0, \quad \forall i, j$ – primal feasibility
- ★ $\lambda_i^* \geq 0, \quad \forall i$ – dual feasibility
- ★ $\lambda_i^* f_i(x^*) = 0, \quad \forall i$ – complementary slackness
- ★ $\nabla f_0(x^*) + \sum_i \lambda_i^* \nabla f_i(x^*) + \sum_j \mu_j^* \nabla h_j(x^*) = 0$ – stationarity

CONVEXITY:

KKT conditions $\iff$ Optimality with zero duality gap

Example: equality-constrained quadratic

Fat matrix: $A \in \mathbb{R}^{p \times n}$, $p < n$, $AA^\top$ invertible

$$\begin{array}{ll} \underset{x}{\text{minimize}} & \frac{1}{2} \|x\|^2 \\ \text{subject to} & Ax = b \end{array}$$

■ DUAL FUNCTION

$$\nabla_x L(x, \mu) = x + A^\top \mu = 0 \quad \implies \quad g(\mu) = -\frac{1}{2} \mu^\top AA^\top \mu - b^\top \mu$$

■ DUAL PROBLEM

$$\underset{\mu}{\text{maximize}} \quad g(\mu) \quad \implies \quad \mu^\star = -(AA^\top)^{-1}b$$

KKT: $x^\star = -A^\top \mu^\star = A^\top (AA^\top)^{-1}b$ – least-norm solution

KKT for constrained linear regression

$$\begin{array}{ll}\underset{\theta}{\text{minimize}} & f(\theta) = \|X\theta - y\|^2 \\ \text{subject to} & \theta \geq 0 \\ & \mathbf{1}^\top \theta = 1\end{array}$$

■ KKT CONDITIONS

$$\nabla f(\theta^*) = \lambda^* - \mu^* \mathbf{1}, \quad \lambda^* \geq 0, \quad \lambda_j^* \theta_j^* = 0$$

■ OPTIMALITY

$$\theta_j^* > 0 \quad \implies \quad \frac{\partial f(\theta^*)}{\partial \theta_j} = -\mu^* \leq \frac{\partial f(\theta^*)}{\partial \theta_k}, \quad \forall k$$

KKT recovers the optimality condition over the probability simplex

Equivalent primals can have different duals

■ ORIGINAL PROBLEM

$$\underset{x}{\text{minimize}} \quad \log \left(\sum_{i=1}^m \exp \left(a_i^\top x - b_i \right) \right)$$

★ no constraints – constant dual function $g = p^\star$

■ REFORMULATION

$$\underset{x, y}{\text{minimize}} \quad \log \left(\sum_{i=1}^m \exp(y_i) \right) \quad \text{subject to} \quad y = Ax - b$$

$$\Rightarrow g(\mu) = \begin{cases} -b^\top \mu - \sum_i \mu_i \log \mu_i, & \mathbf{1}^\top \mu = 1, \mu \geq 0, A^\top \mu = 0 \\ -\infty, & \text{otherwise} \end{cases}$$

Trivial reformulation may result in a different dual

Questions?

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